Group 4

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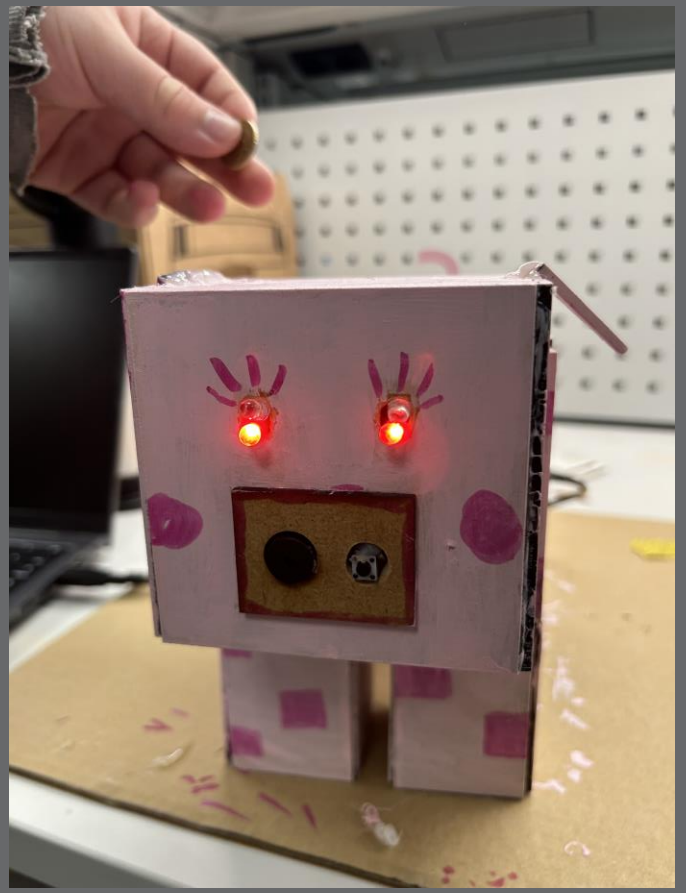
Group 4

Expressive piggy bank



Description and parts used

- The piggy bank gives a happy reaction when you drop money into it through it's coin slot (green eyes, happy sound, tail wag). You can take money from it by opening it's lid, which you can do by pressing a button, but if the secret switch is on it won't open the lid and it will give an angry reaction (red eyes, angry sound, closes the lid).

- We used an IR sensor to the detect coins falling through the coin slot.
- We used two green leds and red leds as eyes. The lid is opened and the tail wags using micro servos. We also used a buzzer to make the happy and angry sounds.

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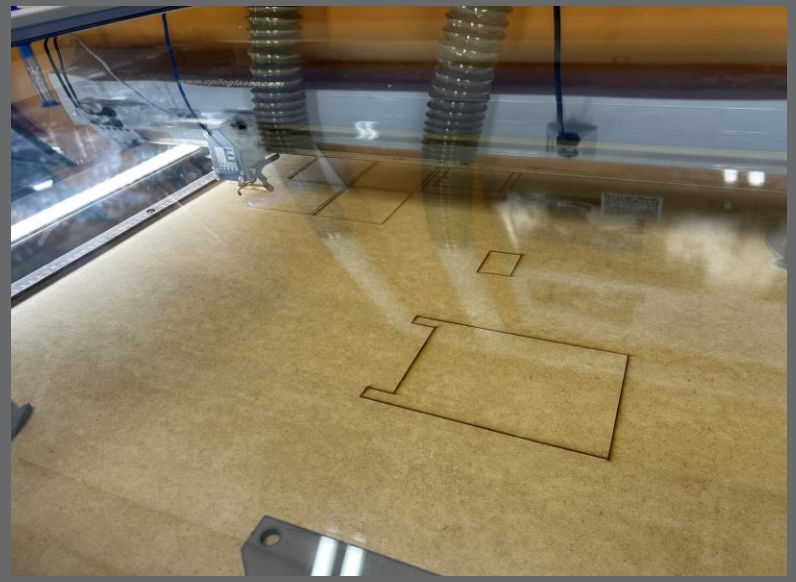
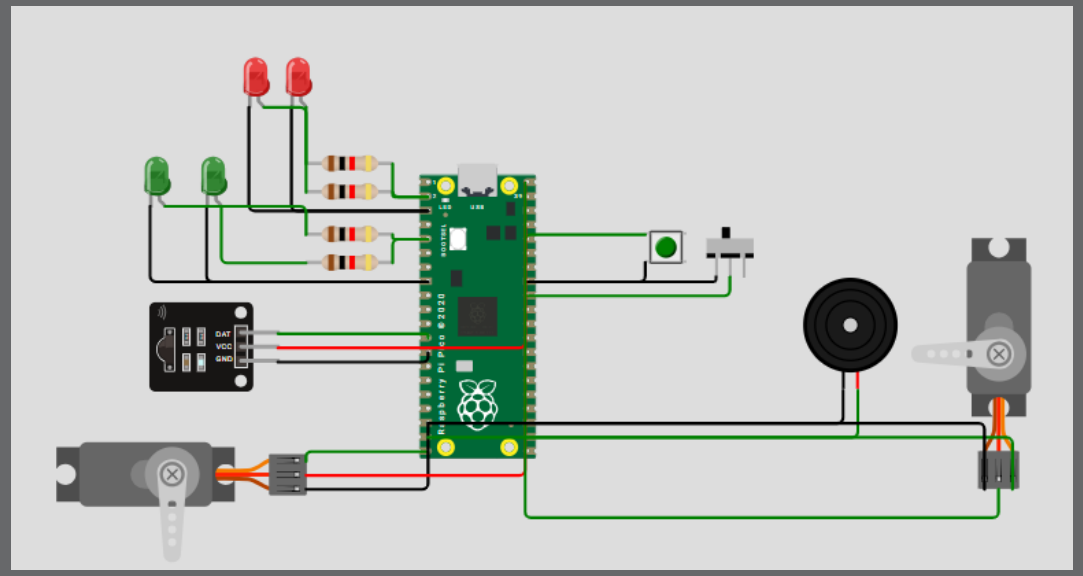


Sustainability

- *It is repairable, but not easily.*
- *We used cardboard inside the pig.*
- *It's back can be taken off, so you can easily look inside of it and repair it.*



Processes

Before laser cutting, we chose the pig model and exported the cutting shapes as a PDF file with specific line color, thickness, and visibility settings so the laser cutter could recognize the cutting paths correctly. We made some changes to the pig and cut it.

Before we began building and connecting the components we created a working simulation and a code for them.

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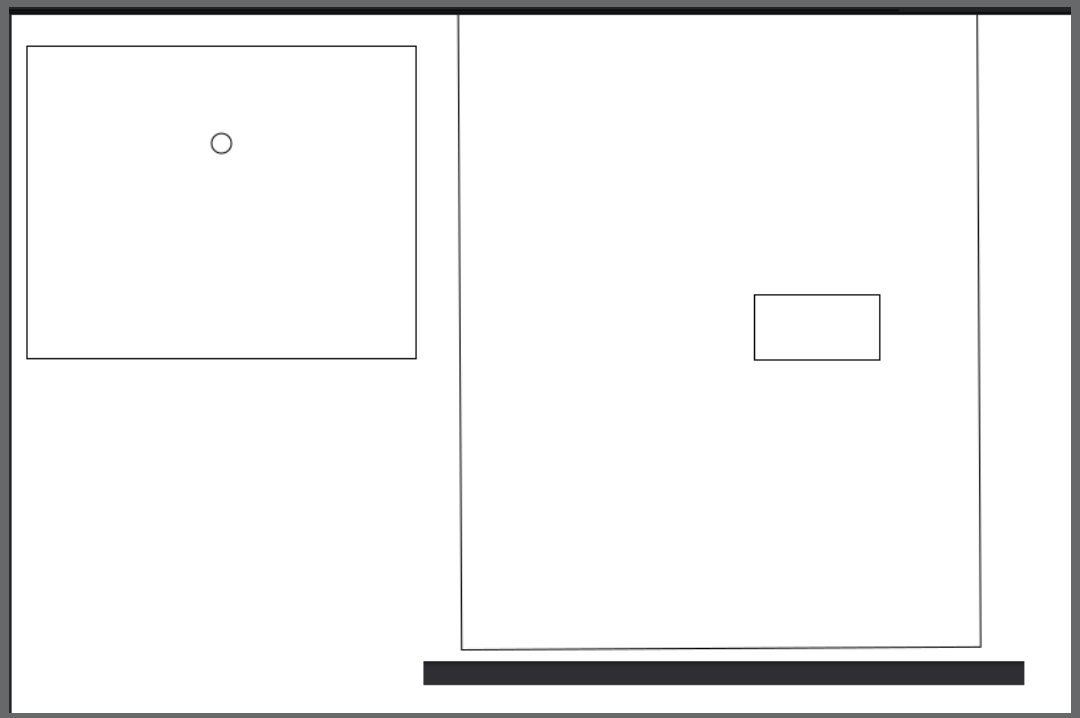

Challenges

- *3D DESIGN: Design failures such as wrong sized holes, not thinking far ahead, assembly mistakes. (Good plan saves a lot of time)*
- *SOFTWARE/ELECTRONICS: We had some challenges keeping all the wires connected.*

Reflection

- *Working as a group went well.*
- *We felt that clearly defined secondary roles were not necessary*
  - *We worked in a self-organizing way*
- *The hardest thing for us was deciding the final design for the piggy bank.*
- *We learned about using new software (Inkscape, wokwi), laser cutting and connecting components.*

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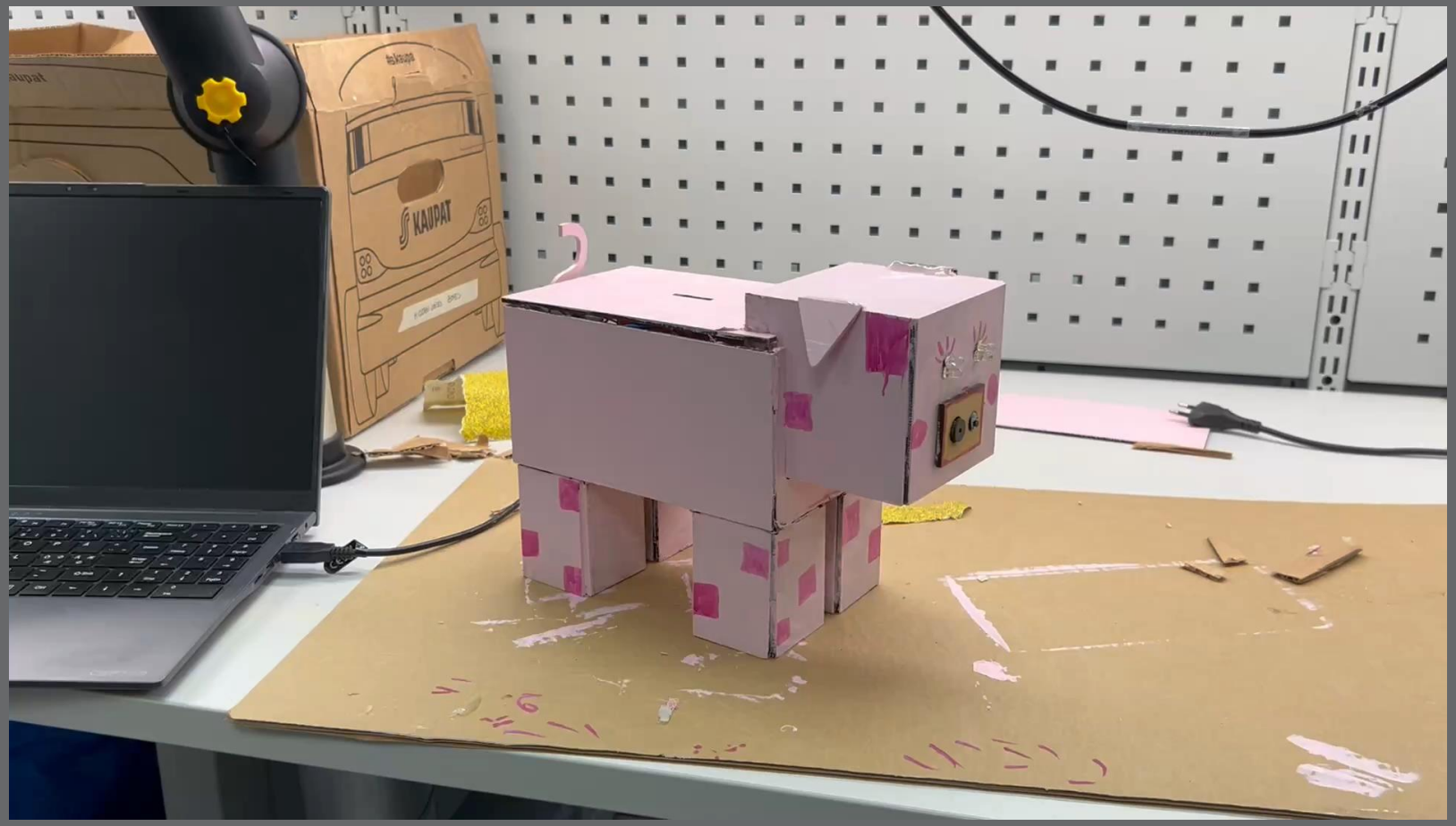
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